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Full Publication List (2022–2026)

PhD students and postdoctoral researchers supervised or co-supervised by F. Picard are shown in bold.

1. F. Picard, V. Rivoirard, A. Roche, and V. M. Panaretos. PCA for point processes. *The Annals of Statistics*, 54(2):929 – 956, 2026
2. M.-D. Ilie, A. Flores-Martinez, M. Chanal, M. Lepetit, **B. Samson**, A. Lehiani, A. Vasiljevic, L. Chinezu, E. Jouanneau, D. Bernard, O. Gandrillon, F. Picard, G. Raverot, and P. Bertolino. Mast cells act as pro-angiogenic and pro-tumorigenic players in pituitary gonadotroph tumors. *Neuro-Oncology*, 28(1):175–190, 2025
3. **R. Belhakem**, F. Picard, V. Rivoirard, and A. Roche. Minimax estimation of functional principal components from noisy discretized functional data. *Scandinavian Journal of Statistics*, 52(1):38–80, 2025
4. M. Ilie, **M. Lepetit**, M. Chanal, A. Vasiljevic, E. Jouanneau, O. Gandrillon, F. Picard, G. Raverot, and P. Bertolino. 8322 mast cells: a new player in gonadotroph tumors. *Journal of the Endocrine Society*, 8(Supplement_1):bvae163–1271, 2024
5. **A. Ozier-Lafontaine**, C. Fourneaux, G. Durif, **P. Arsenteva**, C. Vallot, O. Gandrillon, S. Gonin-Giraud, B. Michel, and F. Picard. Kernel-based testing for single-cell differential analysis. *Genome Biology*, 25(1):114, 2024
6. C. Fourneaux, L. Racine, C. Koering, S. Dussurgey, E. Vallin, A. Moussy, R. Parmentier, F. Brunard, D. Stockholm, L. Modolo, et al. Differentiation is accompanied by a progressive loss in transcriptional memory. *BMC biology*, 22(1):58, 2024
7. C. Arpin, F. Picard, and O. Gandrillon. In silico single-cell metabolism analysis unravels a new transition stage of cd8 t cells 4 days post-infection. *ImmunoInformatics*, page 100038, 2024
8. **M. Lepetit**, M. D. Ilie, M. Chanal, G. Raverot, P. Bertolino, C. Arpin, F. Picard, and O. Gandrillon. scan1. 0: A reproducible and standardized pipeline for processing 10X single cell RNAseq data. *In Silico Biology*, 15(1-2):11–21, 2023
9. J. Poulet-Benedetti, C. Tonnerre-Doncarli, A. L. Valton, M. Laurent, M. Gérard, N. Barinova, N. Parisi, F. Massip, F. Picard, and M. N. Prioleau. Dimeric G-quadruplex motifs-induced NFRs determine strong replication origins in vertebrates. *Nat Commun*, 14(1):4843, Aug 2023
10. **H. Van Assel**, T. Espinasse, J. Chiquet, and F. Picard. A probabilistic graph coupling view of dimension reduction. In *Advances in Neural Information Processing Systems*, volume 35, pages 10696–10708. Curran Associates, Inc., 2022
11. **A. Bonnet**, C. Lacour, F. Picard, and V. Rivoirard. Uniform deconvolution for poisson point processes. *Journal of Machine Learning Research*, 23(194):1–36, 2022
12. S. Zreika, C. Fourneaux, E. Vallin, L. Modolo, R. Seraphin, A. Moussy, E. Ventre, M. Bouvier, **A. Ozier-Lafontaine**, A. Bonnaffoux, et al. Evidence for close molecular proximity between reverting and undifferentiated cells. *BMC biology*, 20(1):1–16, 2022
13. **J. Käfer**, A. Bewick, A. Andres-Robin, G. Lapetoule, A. Harkess, J. Caius, B. Fogliani, G. Gâteblé, P. Ralph, C. W. dePamphilis, et al. A derived ZW chromosome system in amborella trichopoda, representing the sister lineage to all other extant flowering plants. *New Phytologist*, 233(4):1636–1642, 2022

Scientific Software (current)

pppca	Principal Component Analysis for replicated point-process data (R).
ktest	Kernel-based hypothesis testing (R and Python).
KAOV	Kernel hypothesis testing for general experimental designs (Python).
knit	Kernel Normalised Independence Test (Python).
nucount	Methods for estimating absolute cell numbers in spatial transcriptomics spots from histological images (Python).